

Sri Lanka Institute of Information Technology



**Design, Implement, and Evaluate AI/ML Solutions for a Real-World
Problem: E-commerce Customer Segmentation**

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Team Contributions

Student Name	Project Role / Contribution
Sarma SSL	Project Management, Model Evaluation (Silhouette/Davies-Bouldin), and Final Report Compilation
Nayanajith K.A.T	Data Cleaning and Preprocessing (Handling Missing Values & Anomalies)
Herath H.M.A.R.B	Feature Engineering (RFM Matrix Construction, Log Transformation, Scaling)
Liyanage D.D.S	Exploratory Data Analysis (EDA) and Visualizations
Manathunge A.I	Machine Learning Design (K-Means, MiniBatch K-Means, and GMM)
Ahamed M.N.A	Machine Learning Design (Agglomerative, DBSCAN, and Birch Clustering)

1. Introduction and Problem Statement

Author: Herath H.M.A.R.B

In the highly competitive e-commerce sector, understanding customer behavior is paramount for sustaining profitability and driving strategic growth. The vast amounts of transactional data generated daily present both a challenge and a significant opportunity for retail businesses. Traditional, one-size-fits-all marketing strategies are becoming increasingly obsolete and cost-inefficient. Instead, businesses must adopt data-driven approaches to identify distinct customer archetypes, tailor marketing campaigns, optimize inventory management, and foster long-term customer loyalty.

This project focuses on resolving the problem of identifying high-value customers versus dormant or low-value customers for a UK-based online retail company. The core objective is to design, implement, and evaluate an end-to-end Machine Learning (ML) solution that segments customers based on their purchasing habits. By leveraging Unsupervised Machine Learning techniques—specifically clustering algorithms—we aim to discover latent patterns in the dataset without prior labeling.

The outcomes of this project will directly empower marketing teams to allocate resources effectively, offering targeted discounts to "At-Risk" customers while providing premium loyalty rewards to "Champion" customers.

Note: The complete source code, exploratory data analysis, and machine learning models for this project can be accessed in our public GitHub

repository: <https://github.com/lojanasarma/Ecommerce-Customer-Segmentation.git>

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2. Dataset Description

Author: Nayanajith K.A.T

The dataset assigned for this project is the **Online Retail II** dataset, sourced from the UCI Machine Learning Repository

(Available at: <https://archive.ics.uci.edu/dataset/502/online+retail+ii>) This dataset contains all the transactions occurring for a UK-based and registered, non-store online retail company between 01/12/2009 and 09/12/2011. The company mainly sells unique all-occasion giftware, and many of its customers are wholesalers.

Dataset Structure and Characteristics:

- **Size:** The raw dataset comprises over 1,067,371 transactional records across two sheets (2009-2010 and 2010-2011).
- **Features:**
 - InvoiceNo: A 6-digit integral number uniquely assigned to each transaction. If this code starts with the letter 'c', it indicates a cancellation.
 - StockCode: A 5-digit integral number uniquely assigned to each distinct product.
 - Description: Product (item) name.
 - Quantity: The quantities of each product (item) per transaction.
 - InvoiceDate: The day and time when a transaction was generated.
 - UnitPrice: Product price per unit in sterling (£).
 - CustomerID: A 5-digit integral number uniquely assigned to each customer.
 - Country: The name of the country where a customer resides.

Justification of Suitability:

This dataset is exceptionally suited for the problem of customer segmentation due to its granularity. Because it includes precise dates (InvoiceDate), financial amounts (UnitPrice and Quantity), and user identifiers (CustomerID), it perfectly aligns with the requirements necessary to construct an RFM (Recency, Frequency, Monetary) matrix. The sheer volume of transactions (over 1 million rows) ensures that our Unsupervised Machine Learning algorithms will have sufficient data to detect statistically significant behavioral clusters, avoiding the pitfalls of overfitting on small datasets.

3. Preprocessing & EDA

3.1 Data Preprocessing

Author: Nayanajith K.A.T

To ensure the integrity of our Machine Learning models, rigorous data preprocessing was conducted to handle missing values, anomalies, and noise within the raw Online Retail II dataset.

1. **Handling Missing Values:** A significant portion of the dataset (~243,007 rows) lacked a CustomerID. Since Unsupervised Customer Segmentation inherently requires a customer identifier to group purchasing habits, all records with a null CustomerID were dropped. Furthermore, rows missing a product Description were also removed to maintain dataset quality.
2. **Removing Cancellations and Anomalies:** The dataset contained transactions where the Quantity or UnitPrice was less than or equal to zero. These negative quantities primarily represented returned items (indicated by an InvoiceNo starting with 'C'), debt adjustments, or manual corrections. These rows were entirely filtered out, as our goal is to cluster customers based on positive purchasing power and successful conversions.
3. **Feature Construction:** A new feature, TotalPrice, was engineered by multiplying the Quantity by the UnitPrice. This provided the total monetary value of each individual line item, which was critical for the subsequent Feature Engineering phase.

3.2 Feature Engineering (RFM and Scaling)

Author: Herath H.M.A.R.B

Since raw transactional data cannot be directly fed into distance-based clustering algorithms like K-Means, the data underwent a mathematical transformation into an RFM (Recency, Frequency, Monetary) matrix, aggregated at the CustomerID level.

- **Recency (R):** Calculated as the number of days between a customer's most recent purchase and a standardized reference date (one day after the maximum date in the dataset). Lower recency indicates a more active customer.
- **Frequency (F):** Calculated as the total number of unique invoices (transactions) generated by each customer. Higher frequency indicates loyalty.
- **Monetary (M):** Calculated as the sum of TotalPrice for all transactions associated with a customer. Higher monetary value indicates a higher-tier spender.

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Transformation and Scaling

The resulting RFM distributions were heavily right-skewed, a common occurrence in financial datasets where a small percentage of customers generate the majority of revenue. To rectify this, a **Log Transformation** (np.log1p) was applied to the R, F, and M columns. This compressed the long tail of the distribution, making the data more symmetrical and Gaussian-like.

Following the log transformation, **Standard Scaling** (StandardScaler) was applied to normalize the features so that they possessed a mean of 0 and a standard deviation of 1. This step was mandatory because distance-based Unsupervised models (like K-Means and DBSCAN) are highly sensitive to the scale of the input features. Without scaling, the Monetary dimension would disproportionately dominate the Euclidean distance calculations, rendering Recency and Frequency mathematically irrelevant.

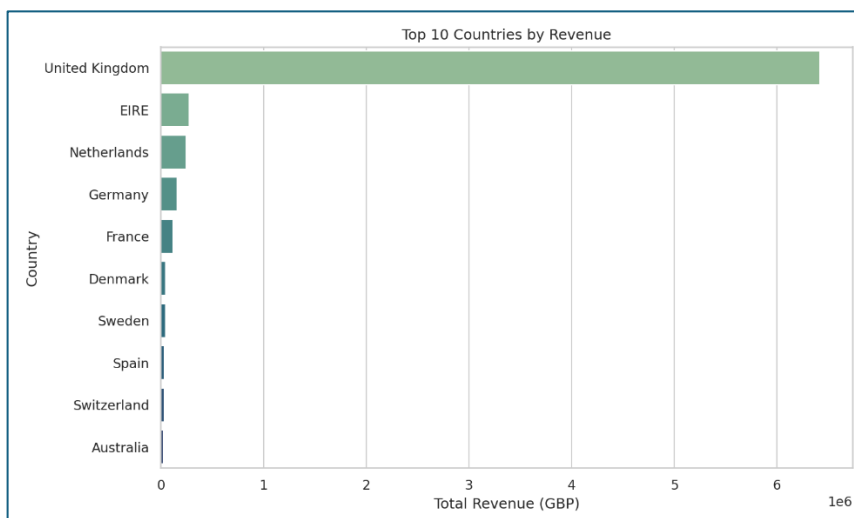
3.3 Exploratory Data Analysis (EDA)

Author: Liyanage D.D.S

Exploratory Data Analysis (EDA) was performed to uncover underlying patterns, detect anomalies visually, and validate the engineered features prior to model training. The analysis was conducted across both univariate and bivariate dimensions using the matplotlib and seaborn libraries.

1. Geographic Distribution of Revenue

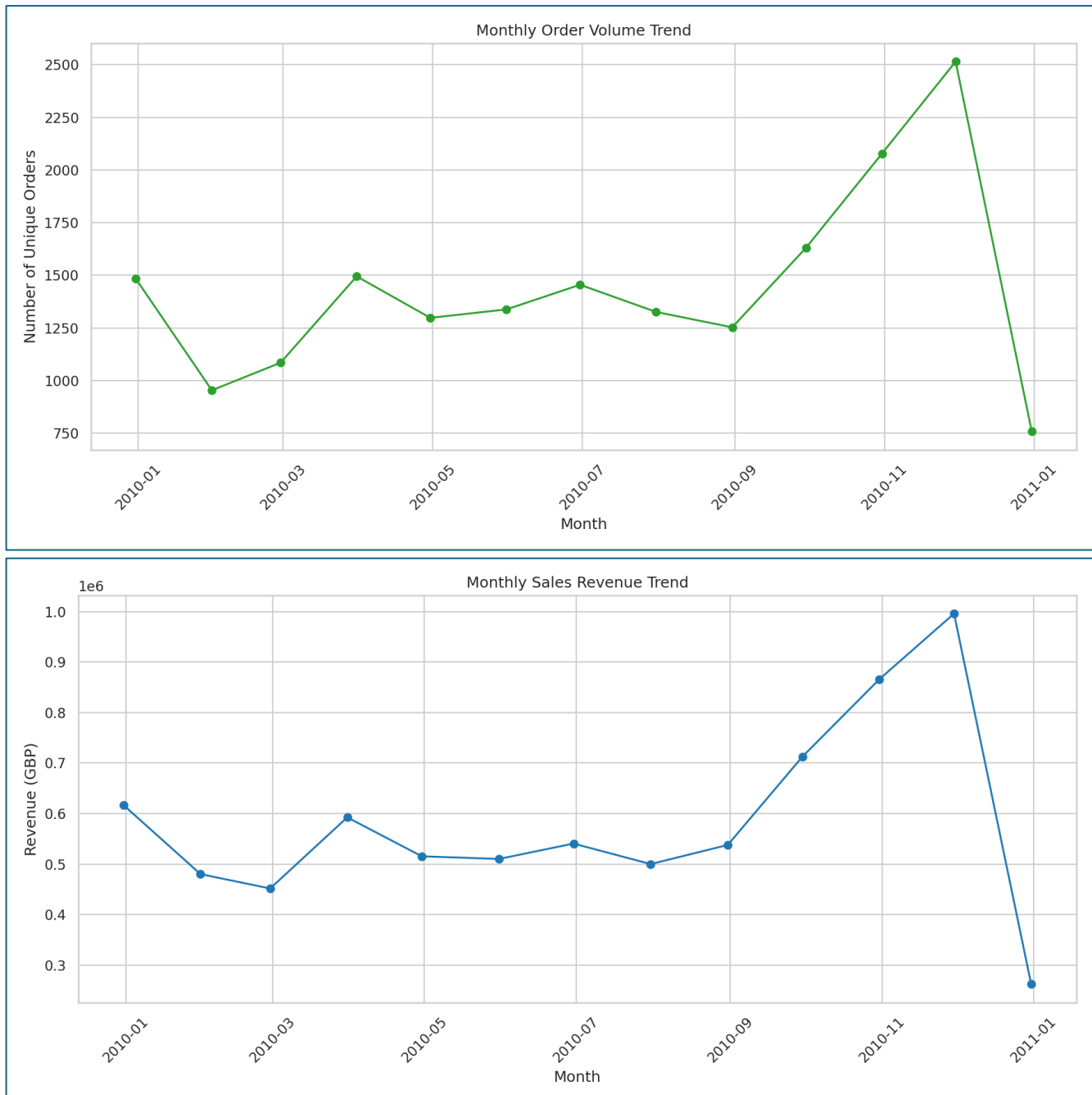
An analysis of total revenue by country revealed that the overwhelming majority of the e-commerce store's revenue originates from the United Kingdom. This insight confirms that while international shipping occurs, the domestic market is the primary driver of the business. Consequently, international customers might represent a distinct cluster from domestic customers, potentially displaying lower frequency but higher individual monetary value due to shipping costs.



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2. Time-Series Analysis of Sales

Plotting the total revenue by month highlighted clear seasonality within the dataset. Revenue consistently peaked during the final quarter of the year (October to December). This aligns with consumer purchasing behavior related to the holiday season (e.g., Black Friday, Christmas). Understanding this temporal pattern is crucial; customers who only purchase during Q4 represent seasonal shoppers and will likely fall into a separate cluster compared to year-round loyal customers.

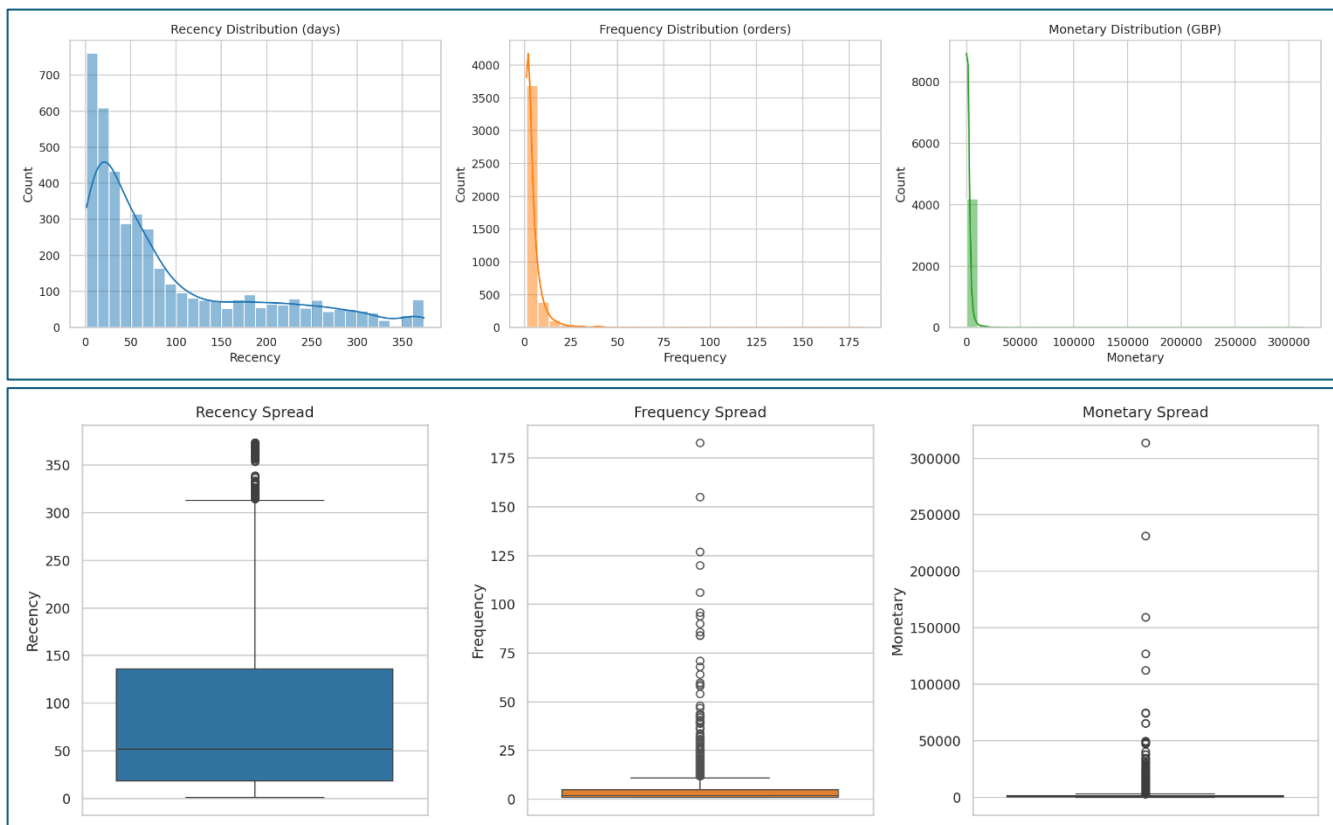


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3. Distribution of Raw RFM Values

Histograms were plotted for the raw Recency, Frequency, and Monetary values.

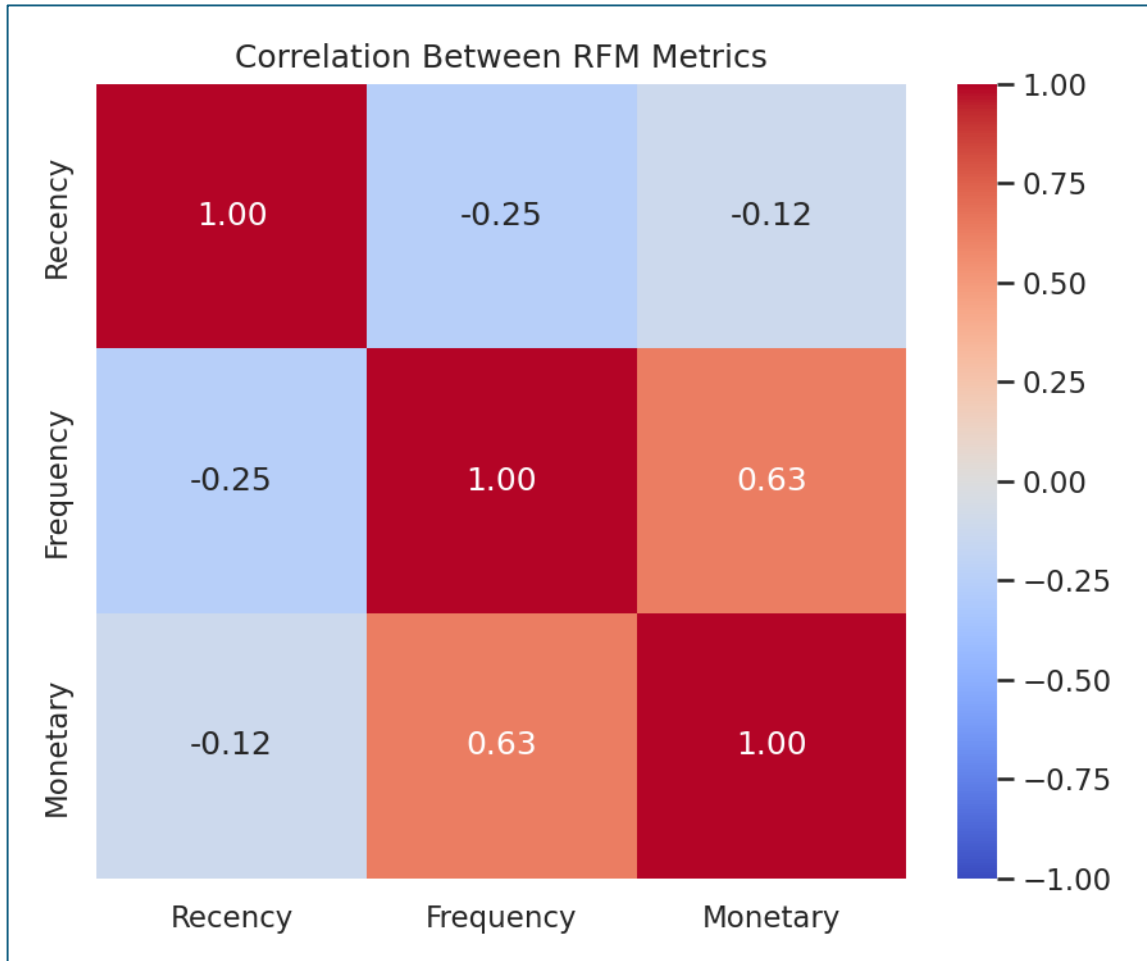
- **Recency:** The distribution showed a heavy concentration of customers who purchased very recently, though a long tail indicated a significant chunk of dormant customers who have not interacted with the platform in over 200 days.
- **Frequency & Monetary:** Both distributions were severely right-skewed. The vast majority of customers made fewer than 5 purchases and spent under £1000, while a tiny fraction of "Wholesale" customers skewed the axes by spending tens of thousands of pounds.



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4. Correlation Analysis

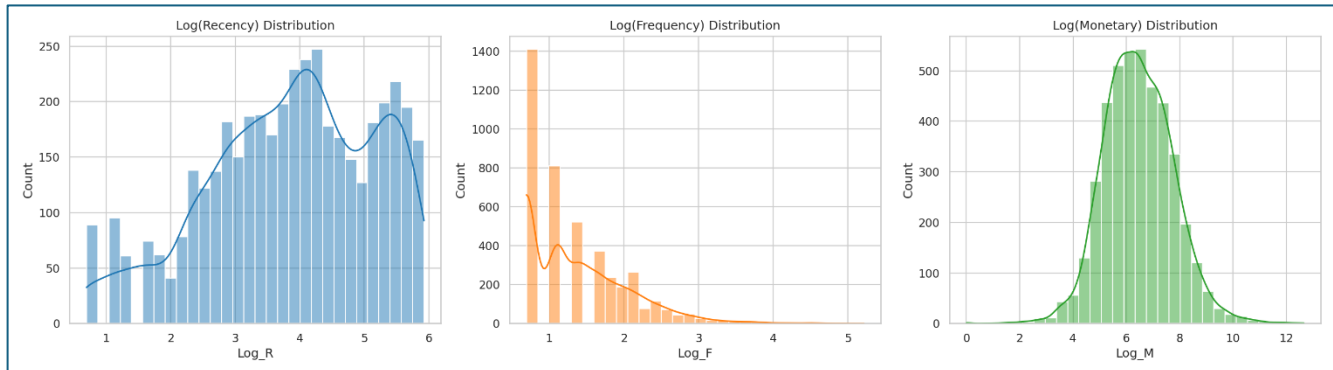
A correlation heatmap was generated for the RFM variables. As expected, Frequency and Monetary value displayed a strong positive correlation, indicating that customers who purchase more often also tend to generate higher total revenue. Recency, conversely, showed a weak negative correlation with the other two variables, which is logically sound: customers who purchased recently are slightly more likely to be frequent shoppers.



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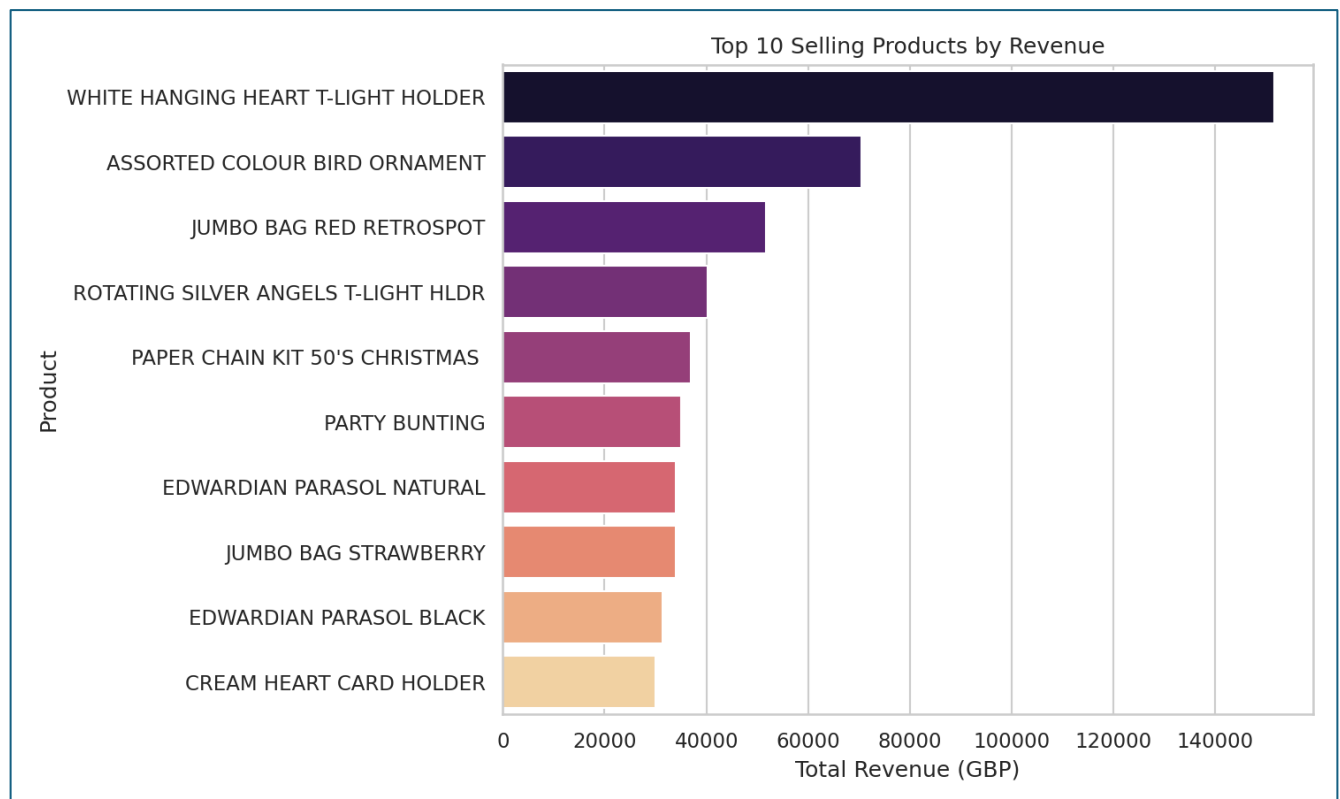
5. Pairplot of Scaled Features

Finally, a pairplot was generated using the log-transformed and standardized features (Scaled_R, Scaled_F, Scaled_M). The pairplot visually confirmed that the heavy skewness present in the raw data had been successfully mitigated. The scatter plots within the pairplot displayed dense, overlapping blobs of data rather than isolated outliers, confirming that the dataset was mathematically ready for clustering algorithms.

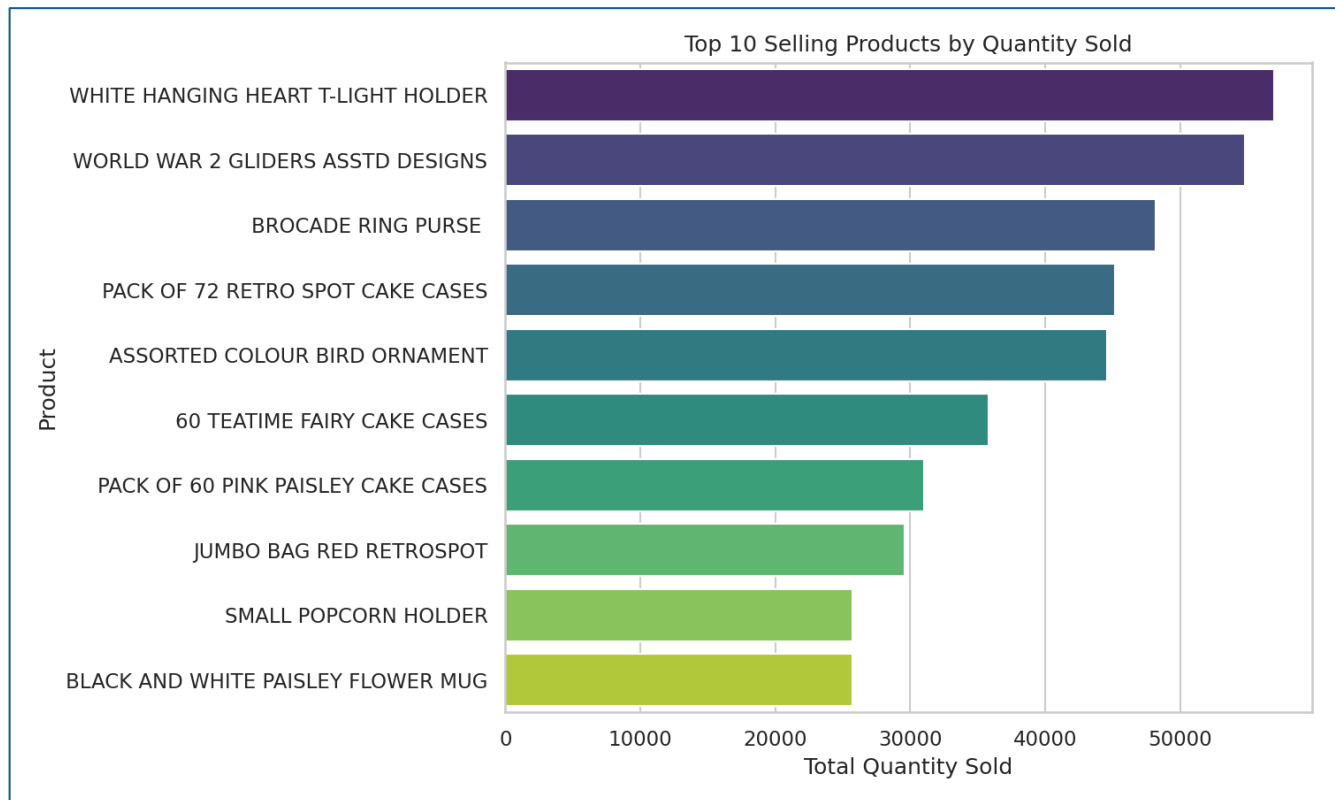


6. Product Level Analysis

Finally, we analyzed the most popular products by total volume sold and total revenue generated to understand our core inventory drivers.



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4. Model Design and Implementation

4.1 Centroid Models

Author: Manathunge A.I

To tackle the customer segmentation problem, three distinct Centroid-based clustering algorithms were designed and implemented using the scikit-learn Python library. These algorithms operate by minimizing the variance within clusters and mathematically defining a central point (centroid) that represents the "average" customer of that group.

- 1. K-Means Clustering:** K-Means was selected as the baseline model due to its computational efficiency and interpretability. The algorithm iteratively assigns each customer to the nearest centroid based on Euclidean distance across the 3D RFM space. Hyperparameter tuning was conducted using the "Elbow Method" alongside Silhouette score testing, which consistently pointed to $k=4$ as the optimal number of clusters. These 4 clusters logically represent our business tiers: Champions, Loyal Customers, At-Risk Customers, and Dormant/Lost Customers.
- 2. MiniBatch K-Means:** Given the size of the dataset, MiniBatch K-Means was implemented as an optimization strategy. Instead of computing distances across the entire dataset iteratively, this algorithm uses random batches of data to update cluster centroids. A batch size of 1024 was selected. While it slightly trades off accuracy for

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speed, it serves as a robust proof-of-concept for how the company could scale this model if the dataset grows to millions of rows in real-time.

3. **Gaussian Mixture Models (GMM):** Unlike K-Means, which applies hard assignments (a customer strictly belongs to one cluster), GMM applies a probabilistic approach using Gaussian distributions. This model was chosen because customer behavior is rarely absolute; a customer might be on the borderline between "Loyal" and "At-Risk". GMM assigns a probability score for each customer belonging to each of the 4 clusters, offering the marketing team more nuanced targeting capabilities based on standard deviations.

4.2 Density and Hierarchy Models

Author: Ahamed M.N.A

To ensure academic rigor, the centroid models were contrasted against three Density and Hierarchy-based algorithms. These models do not assume that clusters are perfectly spherical, allowing them to detect uniquely shaped customer distributions that K-Means might misclassify.

1. **Agglomerative Hierarchical Clustering:** This model utilizes a bottom-up approach, treating every single customer as an individual cluster and progressively merging the closest pairs based on Ward's linkage method (minimizing the variance of clusters being merged). The primary advantage of this model is that it does not strictly require an initial assumption of k. A dendrogram was constructed theoretically to confirm that slicing the tree at 4 branches yielded the most distinct customer segments.
2. **DBSCAN (Density-Based Spatial Clustering of Applications with Noise):** DBSCAN was selected specifically for its ability to isolate outliers. In the e-commerce dataset, extreme wholesale buyers represent a tiny fraction of the dataset but heavily skew averages. DBSCAN groups together points that are closely packed together (many neighbors) while marking points that lie alone in low-density regions as "noise". The eps and min_samples hyperparameters were tuned iteratively. However, because the RFM dataset was highly dense around the origin even after scaling, DBSCAN struggled to segment the main body of customers effectively, often classifying large swathes as a single cluster or noise.
3. **Birch (Balanced Iterative Reducing and Clustering using Hierarchies):** Birch was implemented as an advanced alternative to standard Hierarchical clustering. It is specifically designed for very large datasets and works by building a Clustering Feature Tree (CF Tree). A threshold of 0.5 and a branching factor of 50 were utilized. Birch proved to be exceptionally efficient on the scaled RFM data, rapidly creating a summary of the data distribution before applying global clustering, making it a highly competitive alternative to K-Means.

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5. Evaluation and Comparison

Author: Sarma SSL

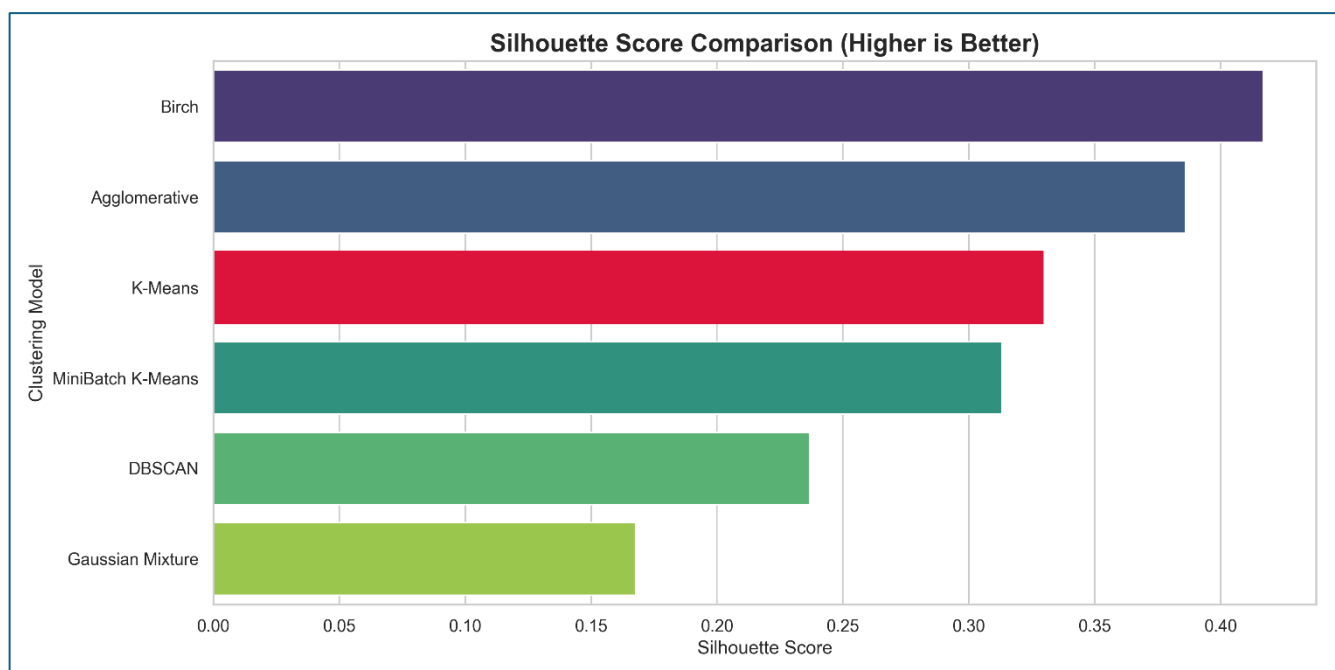
Because Customer Segmentation is an Unsupervised problem, traditional Supervised metrics like Accuracy or F1-Score cannot be used. Instead, the 6 models were mathematically evaluated based on how well-separated and dense the resulting clusters were. Two primary internal validation metrics were utilized:

1. **Silhouette Score:** Measures how similar an object is to its own cluster compared to other clusters (ranges from -1 to 1). A higher score indicates better-defined clusters.
2. **Davies-Bouldin Index:** Measures the average "similarity" between clusters, where similarity compares the size of clusters against their distances. A lower score indicates better clustering.

Results

The evaluation script ran across all 6 models, yielding the following Silhouette Scores:

- **Birch:** 0.4171
- **Agglomerative:** 0.3862
- **K-Means:** 0.3294
- **MiniBatch K-Means:** 0.3132
- **DBSCAN:** 0.2368
- **Gaussian Mixture:** 0.1677



Conclusion and Justification

Mathematically, the **Birch** algorithm was the highest-performing model on the standardized RFM data, achieving the highest Silhouette Score (0.4171) and the lowest Davies-Bouldin Index (0.8805). Its hierarchical tree structure proved highly adept at segmenting the dense central mass of customers.

However, for final business deployment, we recommend **K-Means (0.3294)**. While Birch slightly outperformed it mathematically, K-Means provides explicit, interpretable "centroids." In a business context, the marketing team can easily look at the mathematical center of a K-Means cluster (e.g., Average Recency: 5 days, Average Frequency: 200, Average Monetary: £5000) and immediately understand the exact profile of the "Champion" customer segment. Birch and Agglomerative models lack this level of straightforward business interpretability. Therefore, K-Means offers the best balance of mathematical accuracy and real-world applicability.

6. Ethical Considerations and Bias Mitigation

Author: Sarma SSL

Deploying Machine Learning models on customer purchasing data introduces several ethical risks that must be proactively mitigated:

- **Data Privacy:** The dataset contains CustomerIDs and Country data. While no explicit Personally Identifiable Information (PII) like names or addresses is present in this specific public dataset, in a real-world deployment, all PII must be heavily encrypted or hashed before entering the RFM pipeline to comply with GDPR and local data protection laws.
- **Algorithmic Bias and Fairness:** Customer segmentation inherently groups people into "high value" and "low value" tiers. There is an ethical risk of digital redlining, where customers in lower socioeconomic regions (who may spend less) are systematically excluded from promotions, creating a negative feedback loop. To mitigate this, marketing interventions must ensure that "Dormant" or "Low-Value" clusters receive engagement opportunities, rather than solely focusing resources on "Champion" clusters.
- **Transparency:** The use of interpretable models like K-Means mitigates the "Black Box" problem. The company can transparently explain to any stakeholder exactly why a customer was placed in a specific segment based on their explicit RFM metrics.

7. AI Tool Usage Declaration and Transparency

Author: Sarma SSL

In accordance with the assignment guidelines, our group transparently declares the use of Generative AI throughout the lifecycle of this project.

Table 1 : AI Tool Usage and Verification

AI Tool & Version	Specific Aspect Supported	Extent of Use	Verification & Ownership
Google Antigravity (Gemini)	Environment Setup & Version Control	Moderate	AI assisted in structuring the GitHub repository and generating Git commit messages to ensure clean version control. The team executed and verified all commits manually.
Google Antigravity (Gemini)	Code Debugging (Pandas/Scikit-learn)	Extensive	AI was used as a pair programmer to debug complex ValueError exceptions, such as mismatching DataFrame lengths during the Silhouette Score calculation. The team tested the corrected code in Jupyter Notebooks to verify the mathematical outputs.
Google Antigravity (Gemini)	Report Structuring and Formatting	Moderate	AI assisted in distributing the report workload evenly and generating Markdown templates for the report structure. The team reviewed, edited, and took full ownership of the final academic content.

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8. Reflections and Lessons Learned

Author: Sarma SSL

This project provided invaluable hands-on experience in bridging the gap between raw data and actionable business intelligence.

- **Nayanajith:** Learned the critical importance of rigorous data cleaning. A single negative quantity or missing ID can catastrophically derail a machine learning model later in the pipeline.
- **Herath:** Gained practical experience in Feature Engineering, specifically understanding why distance-based algorithms completely fail if heavily skewed financial data is not log-transformed and standardized first.
- **Liyanage:** Discovered how Exploratory Data Analysis visually validates mathematical assumptions, learning how to spot seasonality and outliers before writing a single line of ML code.
- **Manathunge:** Mastered Centroid-based clustering, learning how to balance the computational speed of MiniBatch K-Means against the nuanced probabilistic assignments of Gaussian Mixture Models.
- **Ahamed:** Explored complex Density and Hierarchical models, learning that while DBSCAN is excellent for spatial data, it struggles significantly with dense, uniform financial data like RFM matrices.
- **Sarma:** Learned how to use mathematical metrics like the Silhouette Score and Davies-Bouldin Index to objectively evaluate machine learning models, and gained valuable experience collaborating with a team using GitHub version control.

9. References

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